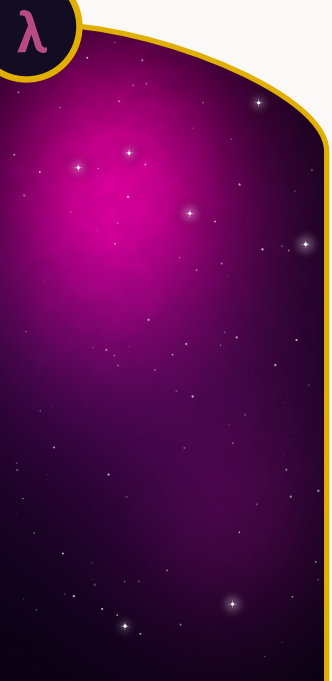


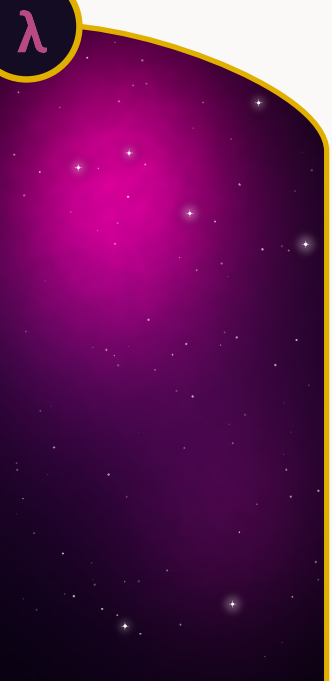
L10A: Searching and Sorting II; Memoization

CS1101S: Programming Methodology

Martin Henz

21st October, 2026

Version 1.1 — Built: 2026-10-20 23:24:53 SGT

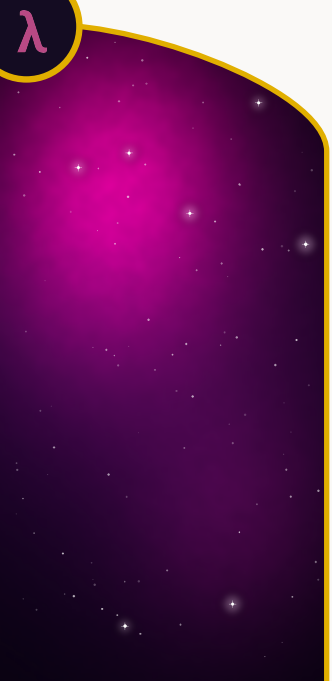


Outline

Searching

Sorting

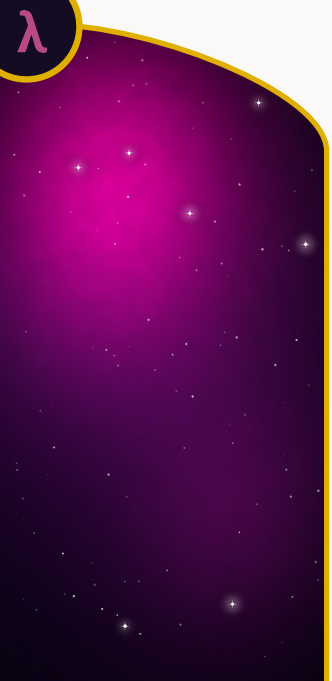
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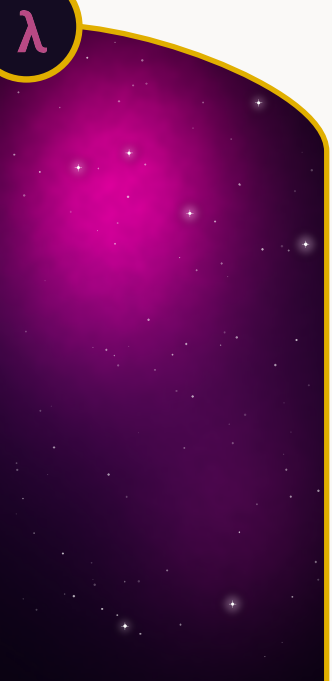
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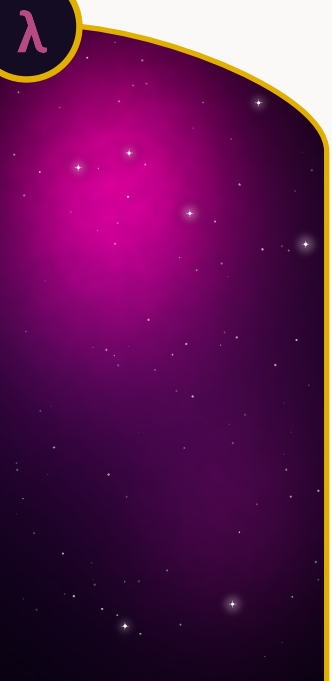


Linear Search



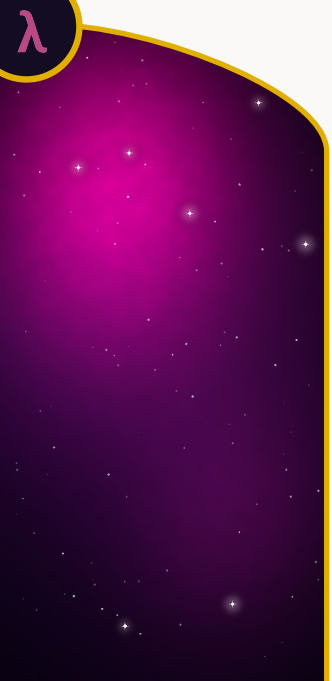
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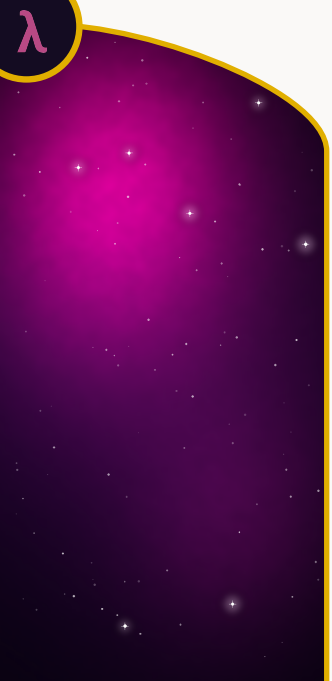
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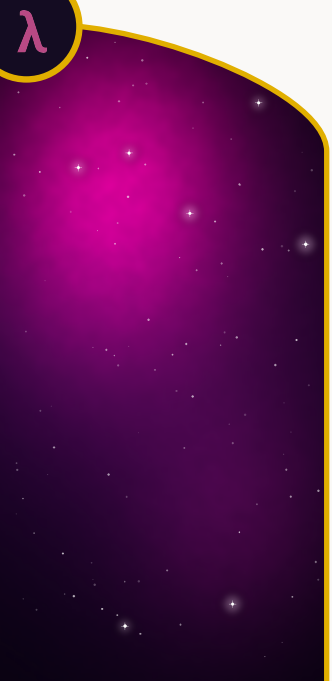
```
def linear_search(A, value):
    i = 0
    while i < len(A) and A[i] != value:
        i = i + 1
    return i < len(A)

print(linear_search([8, 2, 13, 5, 10, 6], 10))
```



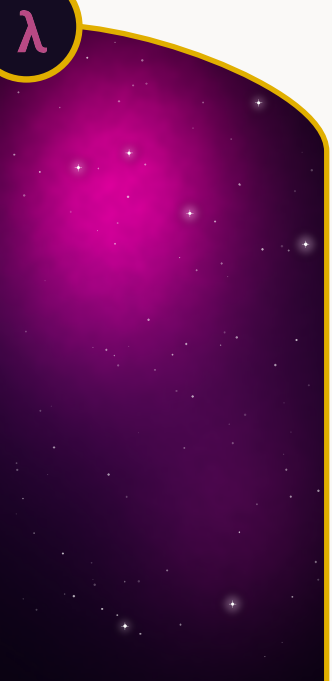


Binary Search



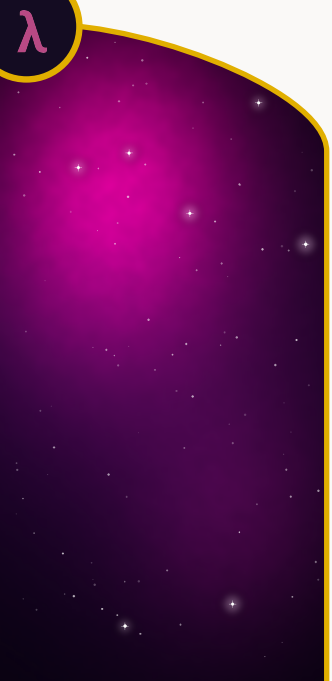
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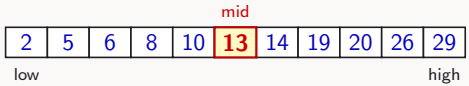


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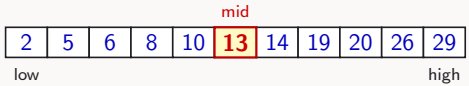


target = 14

$mid = \lfloor (0 + 10)/2 \rfloor = 5$

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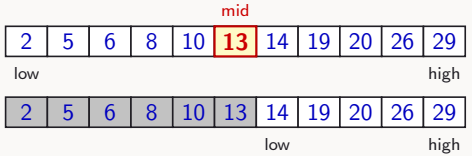
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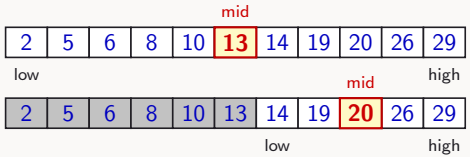
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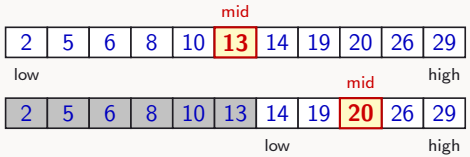
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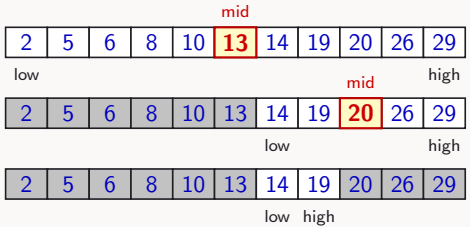
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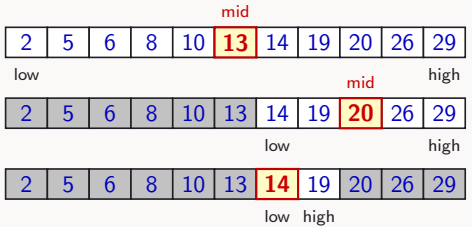
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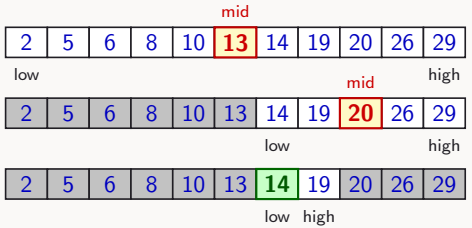
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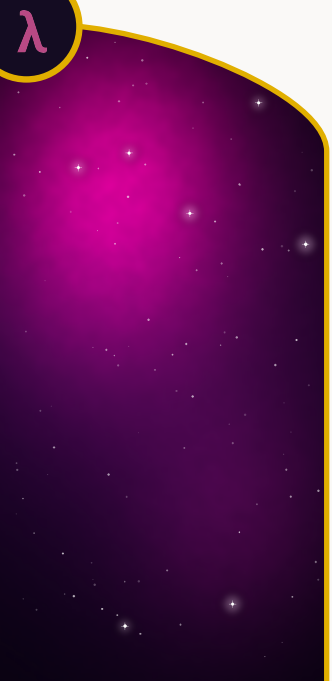


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$mid = \lfloor (6 + 7)/2 \rfloor = 6$
$A[6] = 14 \Rightarrow$ **found**



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- The number of comparisons is $\Theta(\log n)$.
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- A linked list therefore cannot support this binary-search algorithm efficiently.
- We needed a new data structure: binary search trees.

Binary Search Using Recursion

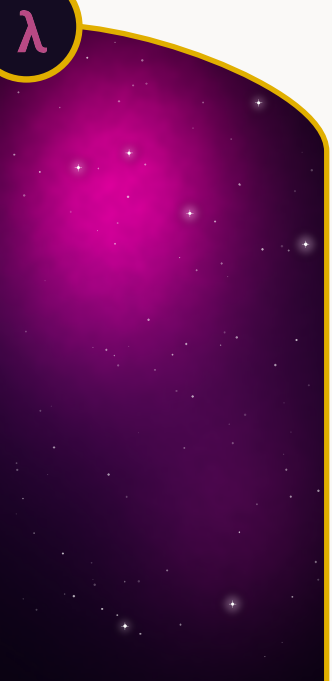
```
def binary_search(A, value):
    def search(low, high):
        if low > high:
            return False
        mid = math_floor((low + high) / 2)
        if value == A[mid]:
            return True
        if value < A[mid]:
            return search(low, mid - 1)
        return search(mid + 1, high)
    return search(0, len(A) - 1)

print(binary_search([2, 5, 6, 8, 10, 13], 8))
```

Binary Search Using a Loop

```
def binary_search(A, value):
    low = 0
    high = len(A) - 1
    while low <= high:
        mid = math_floor((low + high) / 2)
        if value == A[mid]:
            return True
        if value < A[mid]:
            high = mid - 1
        else:
            low = mid + 1
    return False

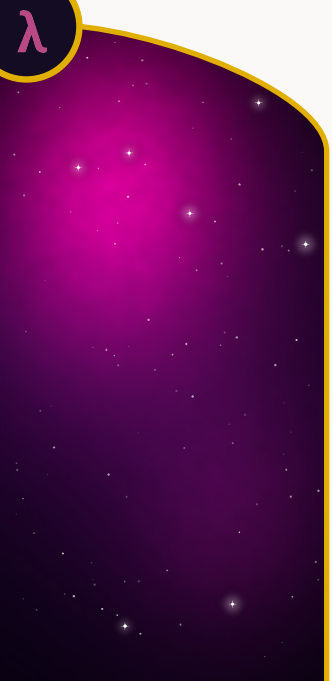
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Check-In time!

<Meet me in the middle.>

Look for Check-In I10A in the Source Academy!



Searching

Sorting

Memoization

Swapping Values Does Not Mutate the List

```
def swap(x, y):
    temp = x
    x = y
    y = temp

A = [4, 5, 6, 7]
swap(A[0], A[3])
print(A)
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- Reassigning `x` and `y` changes only their local bindings.
- The elements of `A` are unchanged.

Swap the List Elements Directly

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def swap(A, i, j):
    temp = A[i]
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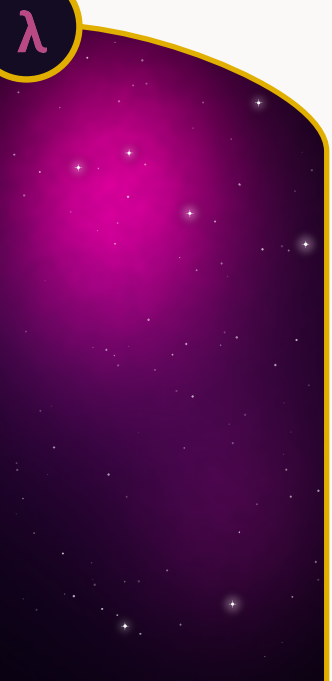
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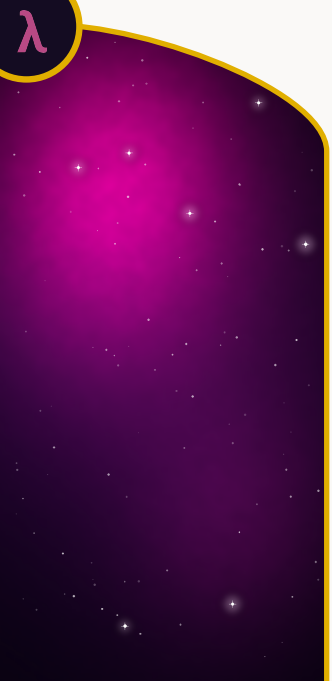
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```



- Assignment to `A[i]` mutates the list itself.

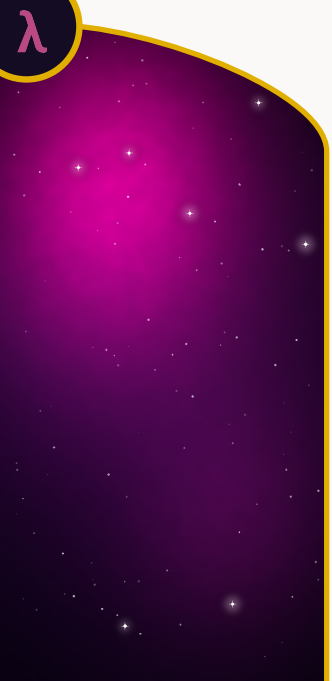


Selection Sort



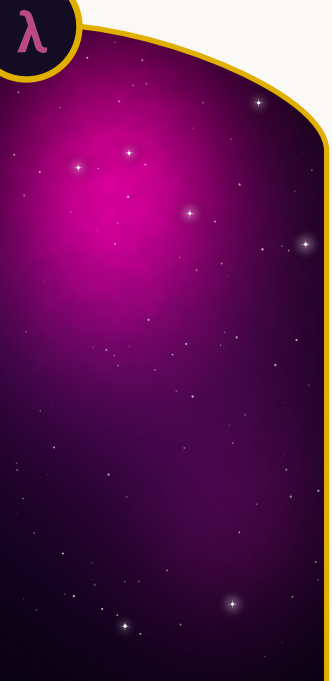
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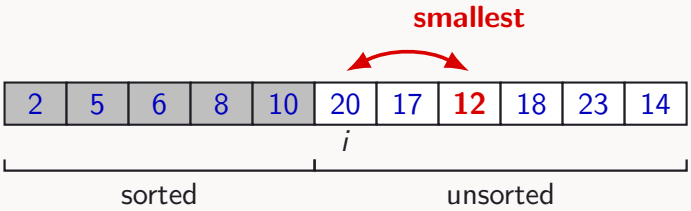


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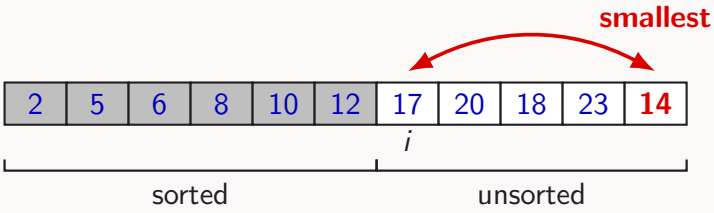
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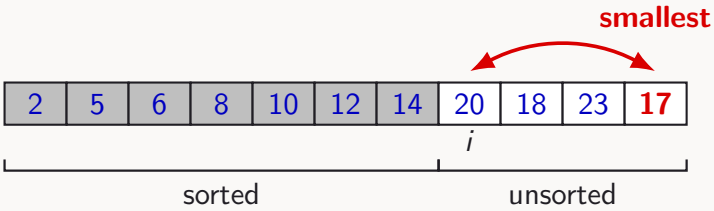
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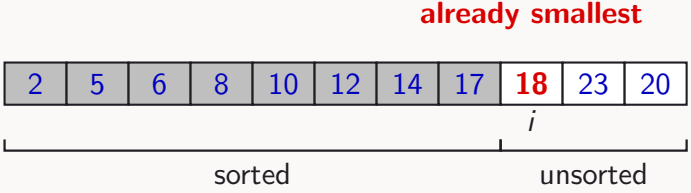
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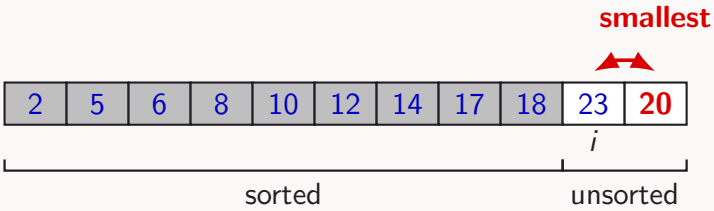
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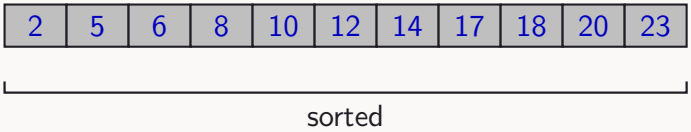
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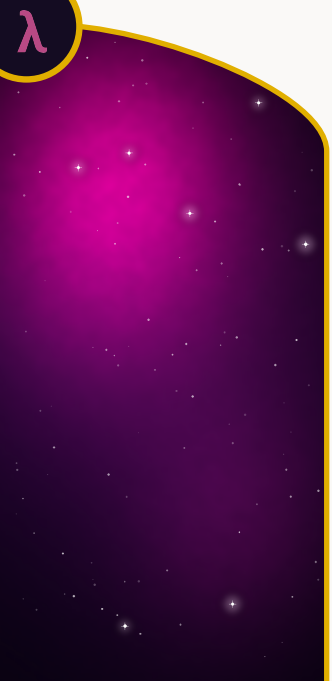
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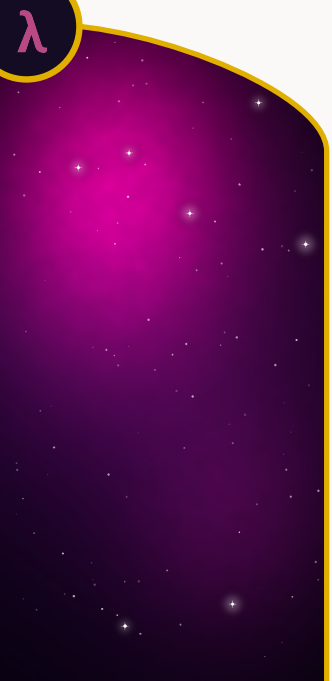
Selection Sort for Lists

```
def selection_sort(A):
    for i in range(len(A) - 1):
        min_pos = i
        for j in range(i + 1, len(A)):
            if A[j] < A[min_pos]:
                min_pos = j
        swap(A, i, min_pos)

A = [10, 5, 8, 2, 6]
selection_sort(A)
print(A)
```

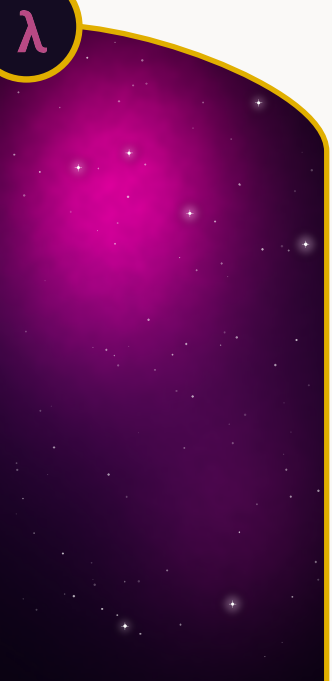


Insertion Sort



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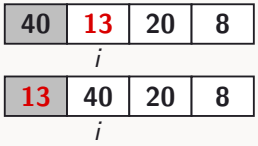


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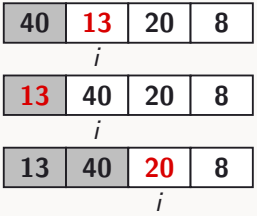
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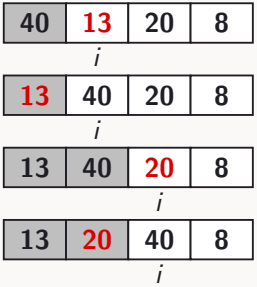
Insertion Sort

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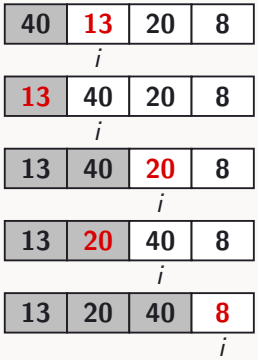
Insertion Sort

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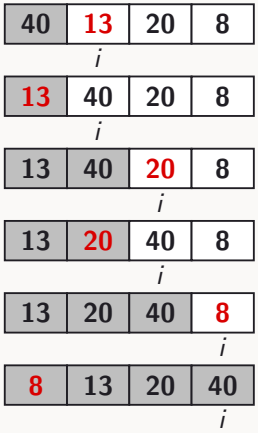
Insertion Sort

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Insertion Sort

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Insertion Sort by Swapping



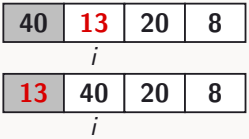

```
def insertion_sort(A):
    for i in range(1, len(A)):
        j = i
        while (j > 0 and
               A[j - 1] > A[j]):
            swap(A, j - 1, j)
            j = j - 1

A = [40, 13, 20, 8]
insertion_sort(A)
print(A)
```

Insertion Sort by Swapping

```
def insertion_sort(A):
    for i in range(1, len(A)):
        j = i
        while (j > 0 and
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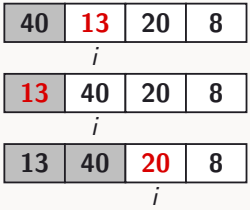
A = [40, 13, 20, 8]
insertion_sort(A)
print(A)
```



Insertion Sort by Swapping

```
def insertion_sort(A):
    for i in range(1, len(A)):
        j = i
        while (j > 0 and
               A[j - 1] > A[j]):
            swap(A, j - 1, j)
            j = j - 1

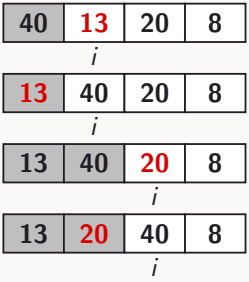
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```



Insertion Sort by Swapping

```
def insertion_sort(A):
    for i in range(1, len(A)):
        j = i
        while (j > 0 and
               A[j - 1] > A[j]):
            swap(A, j - 1, j)
            j = j - 1

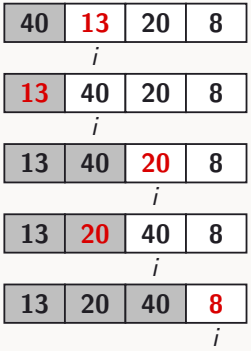
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Insertion Sort by Swapping

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def insertion_sort(A):
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        j = i
        while (j > 0 and
               A[j - 1] > A[j]):
            swap(A, j - 1, j)
            j = j - 1

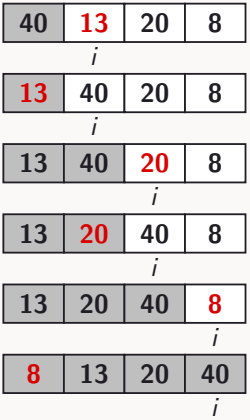
A = [40, 13, 20, 8]
insertion_sort(A)
print(A)
```



Insertion Sort by Swapping

```
def insertion_sort(A):
    for i in range(1, len(A)):
        j = i
        while (j > 0 and
               A[j - 1] > A[j]):
            swap(A, j - 1, j)
            j = j - 1

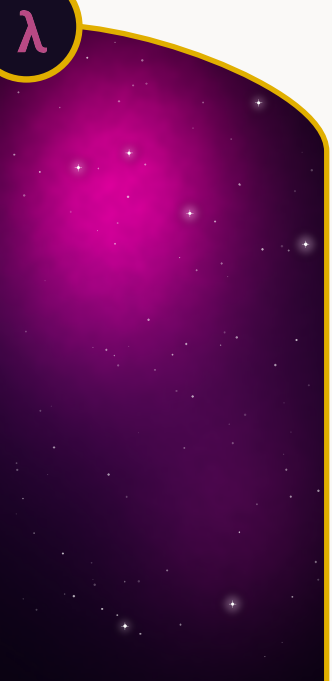
A = [40, 13, 20, 8]
insertion_sort(A)
print(A)
```



Insertion Sort by Shifting

```
def insertion_sort(A):
    for i in range(1, len(A)):
        value = A[i]
        j = i - 1
        while j >= 0 and A[j] > value:
            A[j + 1] = A[j]
            j = j - 1
        A[j + 1] = value

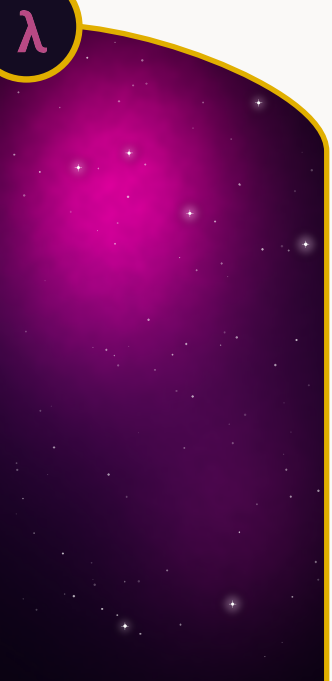
A = [40, 13, 20, 8]
insertion_sort(A)
print(A)
```



Merge Sort

Merge Sort

- Split the list into halves until each part has one element.

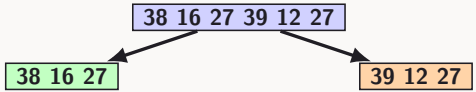


Merge Sort

- Split the list into halves until each part has one element.
- Merge adjacent sorted parts back together.

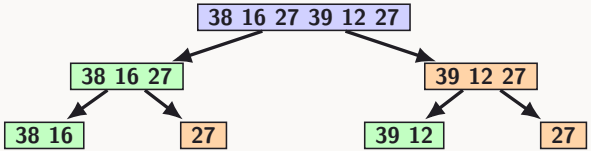
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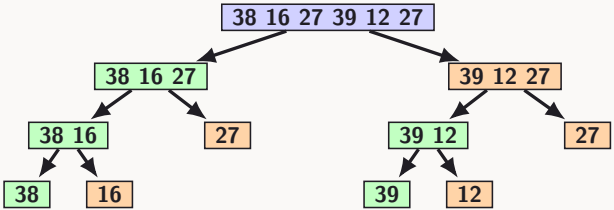
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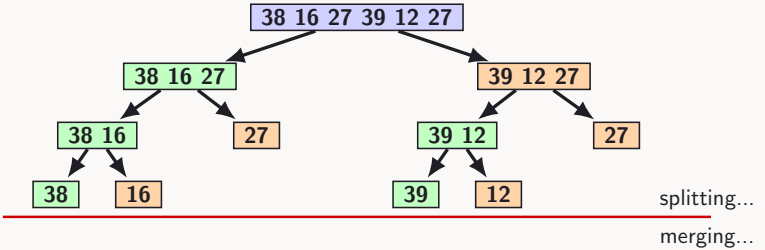
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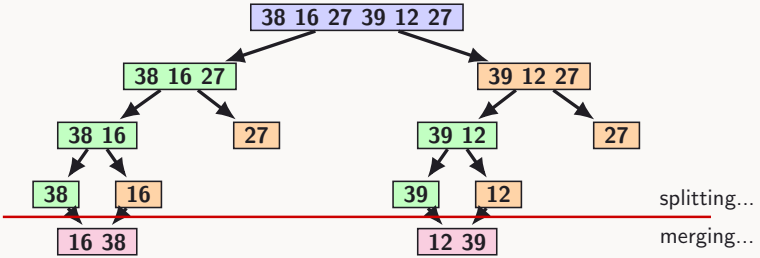
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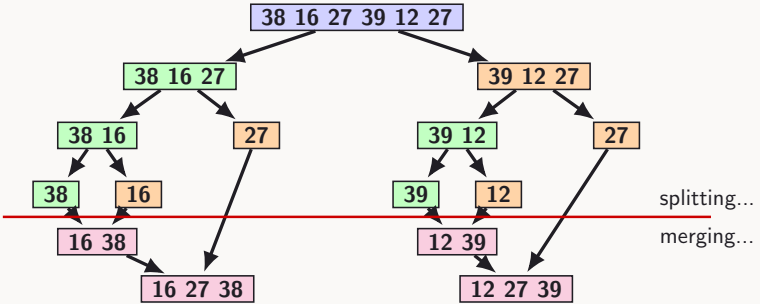
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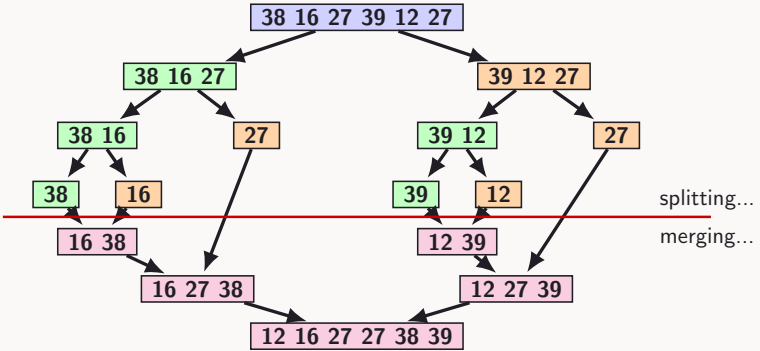
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Merge Sort

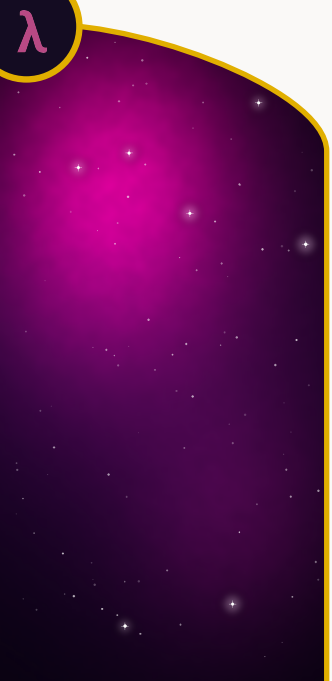
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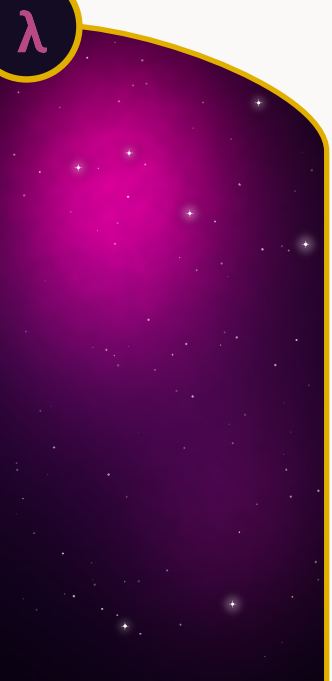
Merge Sort for Lists

```
def merge_sort(A):
    def sort(low, high):
        if low < high:
            mid = math_floor((low + high) / 2)
            sort(low, mid)
            sort(mid + 1, high)
            merge(A, low, mid, high)
    sort(0, len(A) - 1)

A = [5, 2, 8, 3, 7, 4]
merge_sort(A)
print(A)
```



Merging Two Sorted Regions



Merging Two Sorted Regions

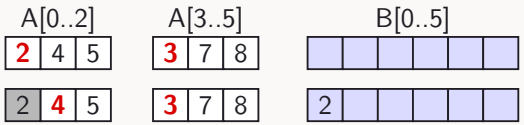
- Repeatedly copy the smaller front element into a temporary buffer.

Merging Two Sorted Regions

- Repeatedly copy the smaller front element into a temporary buffer.
- Copy the completed buffer back into the original list.

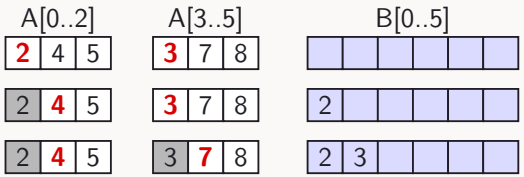
Merging Two Sorted Regions

- Repeatedly copy the smaller front element into a temporary buffer.
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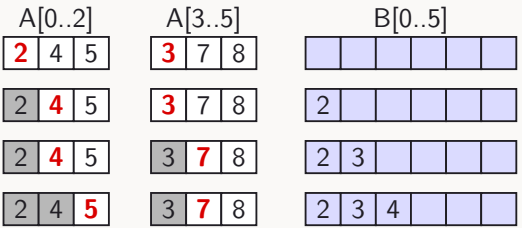
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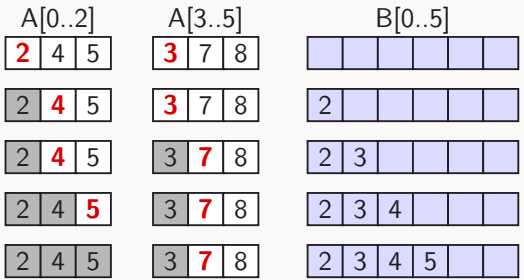
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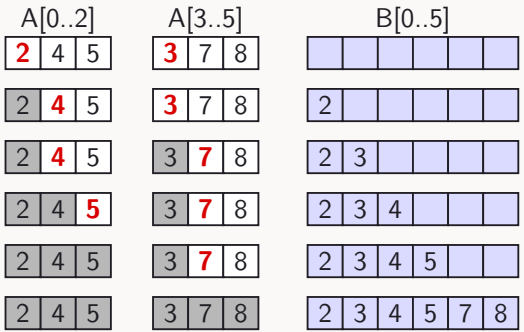
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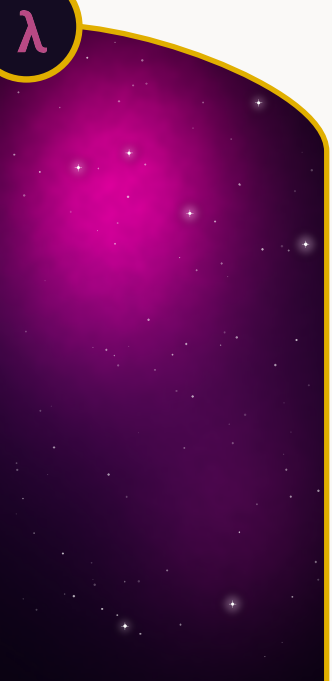
Merging Two Sorted Regions

- Repeatedly copy the smaller front element into a temporary buffer.
- Copy the completed buffer back into the original list.



The Merge Operation

```
def merge(A, low, mid, high):
    B = [None] * (high - low + 1)
    left = low
    right = mid + 1
    i = 0
    while left <= mid or right <= high:
        take_left = right > high or (
            left <= mid and A[left] <= A[right])
        if take_left:
            B[i] = A[left]
            left = left + 1
        else:
            B[i] = A[right]
            right = right + 1
        i = i + 1
    for i in range(len(B)):
        A[low + i] = B[i]
```



Searching

Sorting

Memoization

Naive Fibonacci Repeats Work

```
def fib(n):
    return n if n <= 1 else fib(n - 1) + fib(n - 2)

print(fib(5))
```

Naive Fibonacci Repeats Work

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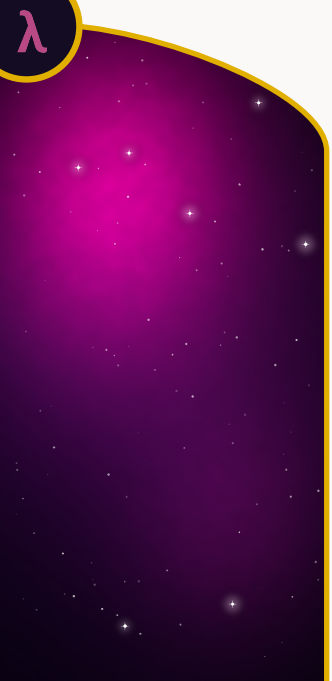
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Naive Fibonacci Repeats Work

```
def fib(n):
    return n if n <= 1 else fib(n - 1) + fib(n - 2)

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```

- Time: $\Theta(\varphi^n)$, where φ is the golden ratio.



Why Exponential?

- The same function calls are evaluated repeatedly.

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fib(5)

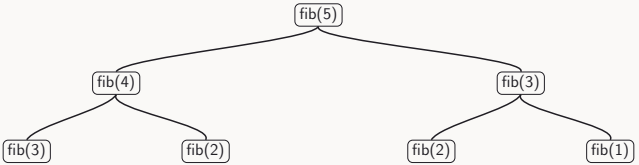
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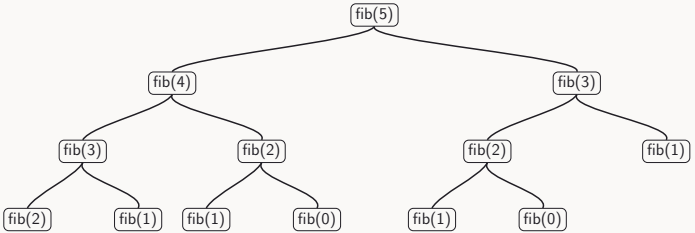
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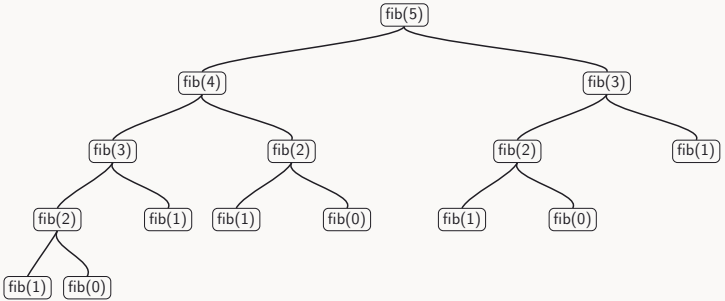
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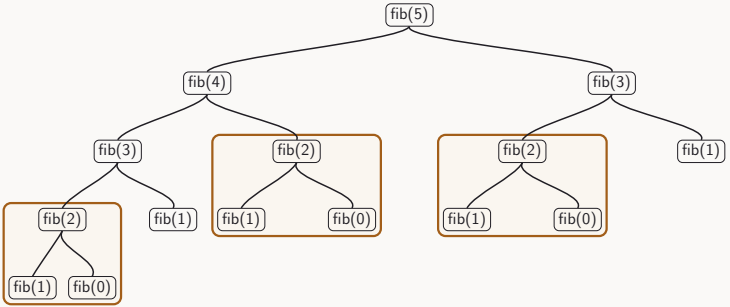
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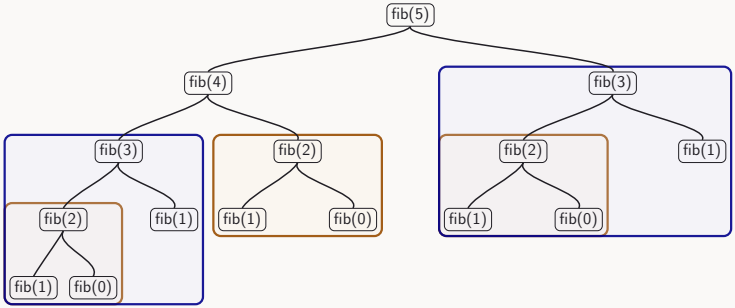
Why Exponential?

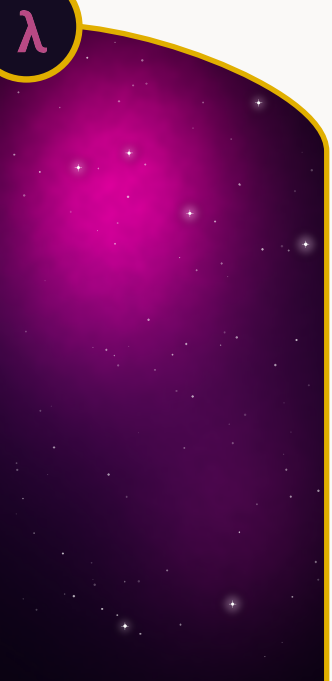
- The same function calls are evaluated repeatedly.



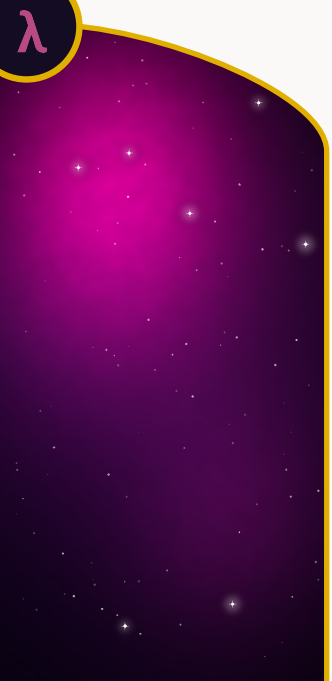
Why Exponential?

- The same function calls are evaluated repeatedly.
- How can we avoid this repeated work?





Memoization



Memoization

- Record a result the first time it is computed.

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- On later calls with the same argument, return the recorded result.

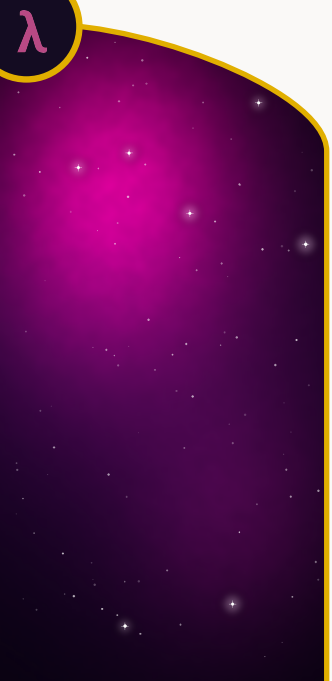
Memoization

- Record a result the first time it is computed.
- On later calls with the same argument, return the recorded result.
- For nonnegative integer arguments, a list gives $\Theta(1)$ lookup.

Memoized Fibonacci with Local Memory

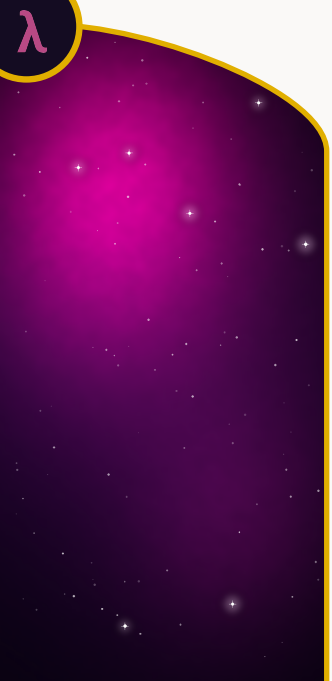
```
def mfib(n):
    mem = [None] * (n + 1)
    def fib(k):
        if mem[k] is not None:
            return mem[k]
        result = k if k <= 1 else fib(k - 1) + fib(k - 2)
        mem[k] = result
        return result
    return fib(n)

print(mfib(10))
```



Evaluating `mfib(5)`

- Each distinct argument is fully evaluated only once.



Evaluating `mfib(5)`

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fib(5)

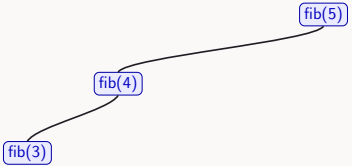
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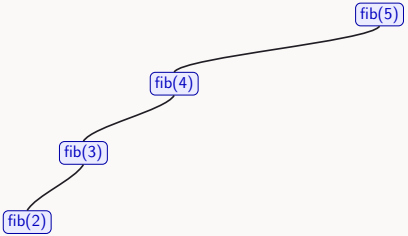
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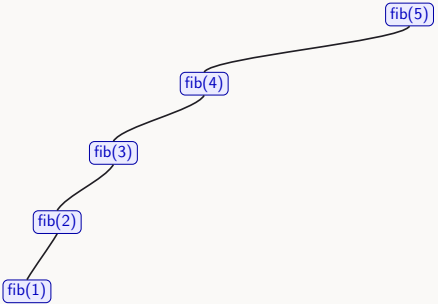
Evaluating `mfib(5)`

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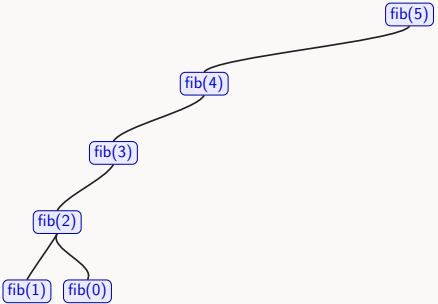
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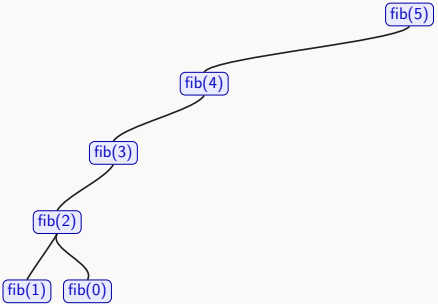
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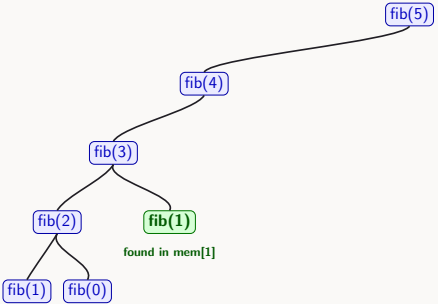
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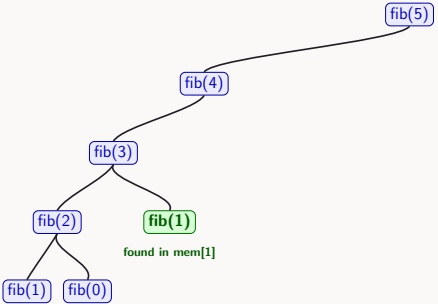
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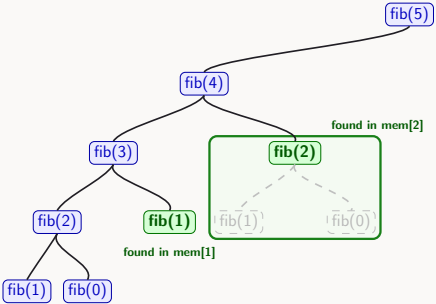
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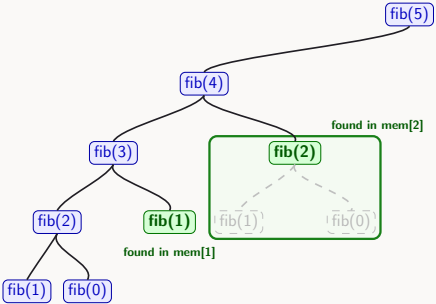
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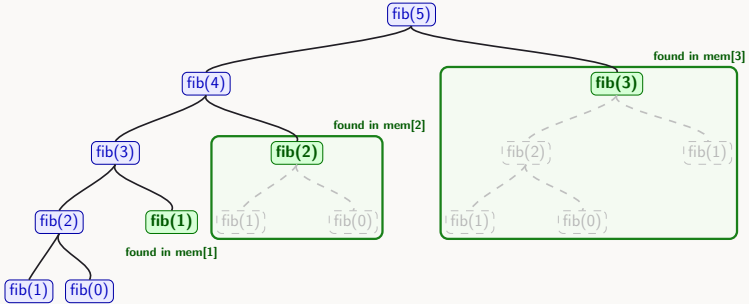
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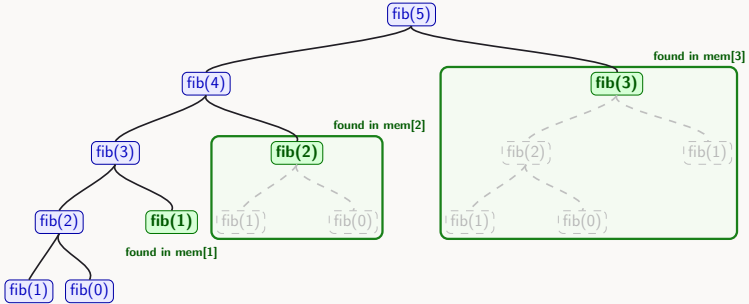
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Evaluating `mfib(5)`

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- Time: $\Theta(n)$; additional space: $\Theta(n)$.

An Abstraction for Memoization



```
def memoize(f, size):
    mem = [None] * size
    def memoized(x):
        if mem[x] is not None:
            return mem[x]
        result = f(x)
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- The returned function retains its private memory between calls.

First Attempt: Memoize Only the Outer Call

```
def fib(n):
    return n if n <= 1 else fib(n - 1) + fib(n - 2)

memo_fib = memoize(fib, 24)
print(memo_fib(23))
```

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- This is still exponential: recursive calls invoke `fib`, bypassing `memo_fib`.

Second Attempt: Memoized Recursion

```
mfib = memoize(
    lambda n: n if n <= 1
              else mfib(n - 1) + mfib(n - 2),
    32)

print(mfib(31))
```

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- Every recursive call now passes through the memoized function.

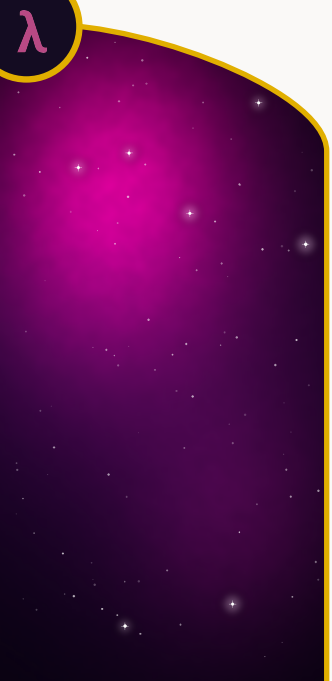
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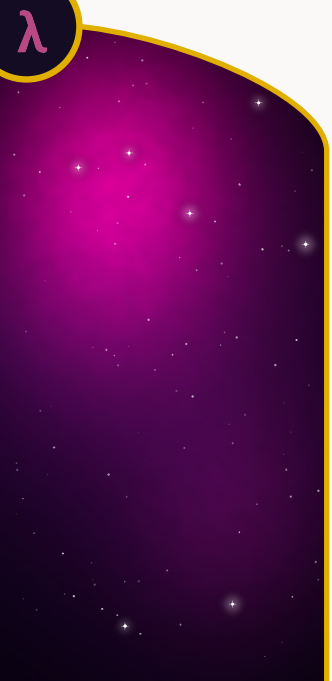
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- Every recursive call now passes through the memoized function.
- Time and additional space are both $\Theta(n)$.

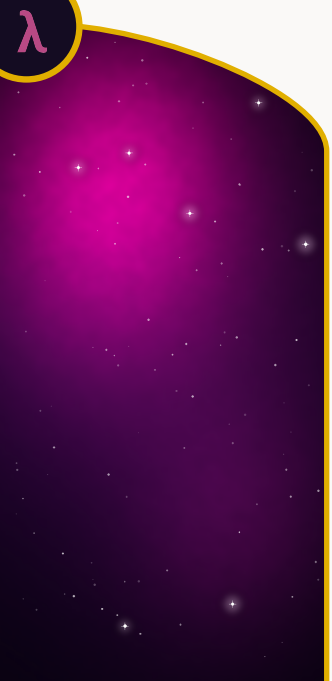


The n-Choose-k Problem



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- How many ways can we choose k items from n items?

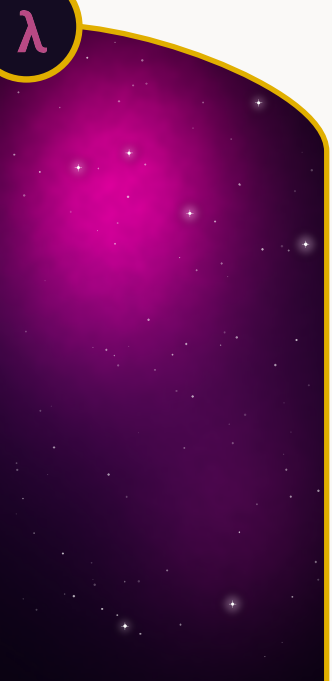


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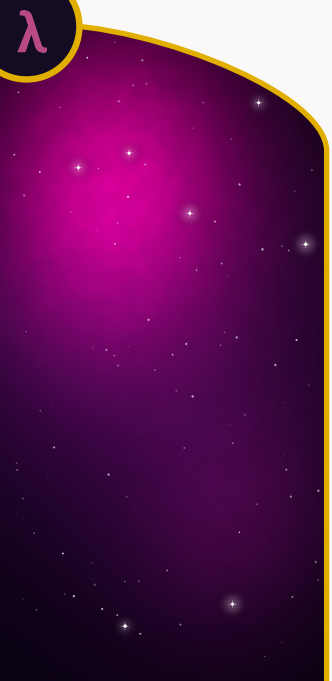
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- How many ways can we choose k items from n items?
- Pick one item:
  - Do not choose it: choose k from the remaining $n-1$.
  - Choose it: choose $k-1$ from the remaining $n-1$.
- Therefore,

$$C(n,k) = C(n-1,k) + C(n-1,k-1).$$

A Straightforward Implementation



```
def choose(n, k):
    if k > n:
        return 0
    if k == 0 or k == n:
        return 1
    return choose(n - 1, k) + choose(n - 1, k - 1)

print(choose(5, 2))
```

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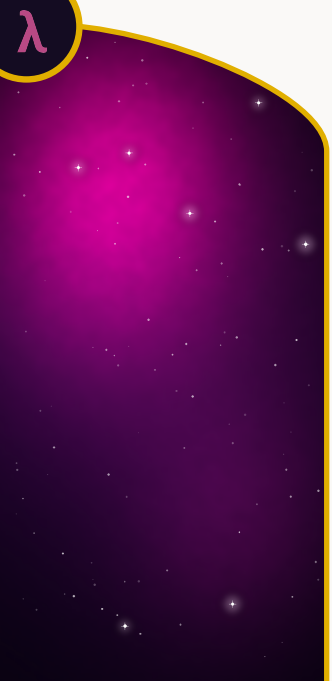
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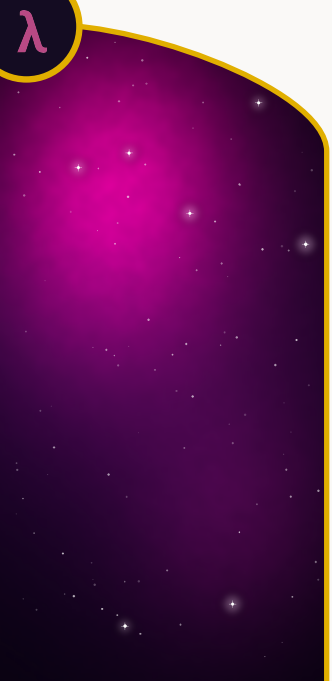
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print(choose(5, 2))
```

- Many pairs (n, k) are evaluated repeatedly.



Memoizing Two Arguments



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- The result depends on the pair (n, k).
- For one call `choose(n, k)`, all relevant arguments satisfy $0 \leq i \leq n$ and $0 \leq j \leq k$.
- A preallocated $(n + 1) \times (k + 1)$ table therefore gives $\Theta(1)$ lookup and update.

Creating a Two-Dimensional Table

```
def make_table(rows, columns):
    table = [None] * rows
    for i in range(rows):
        table[i] = [None] * columns
    return table
```



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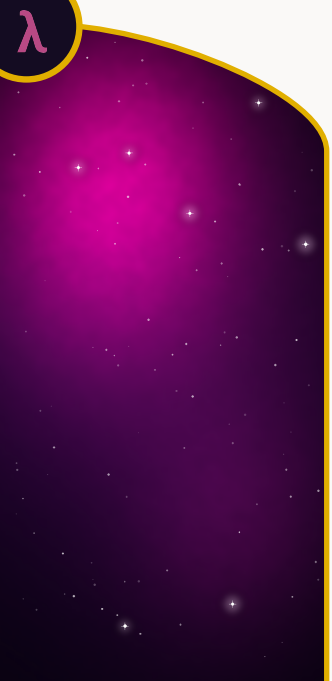


- Each row is a distinct list, so updating one row does not change another.

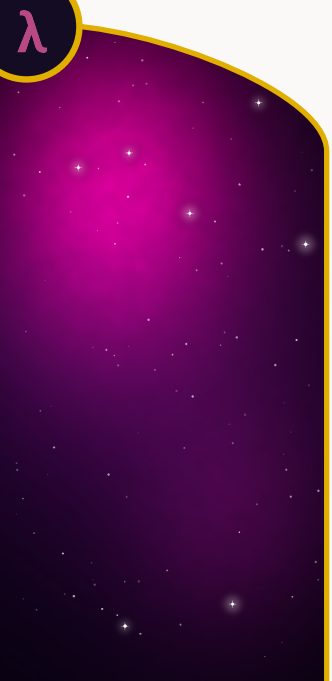
Memoized *n*-Choose-*k*

```
def mchoose(n, k):
    mem = make_table(n + 1, k + 1)
    def choose(i, j):
        if j > i:
            return 0
        if mem[i][j] is not None:
            return mem[i][j]
        result = 1 if j == 0 or j == i else (
            choose(i - 1, j) + choose(i - 1, j - 1))
        mem[i][j] = result
        return result
    return choose(n, k)

print(mchoose(5, 2))
```



Memoized n-Choose-k: Cost



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- Each reachable pair (i, j) is computed at most once.

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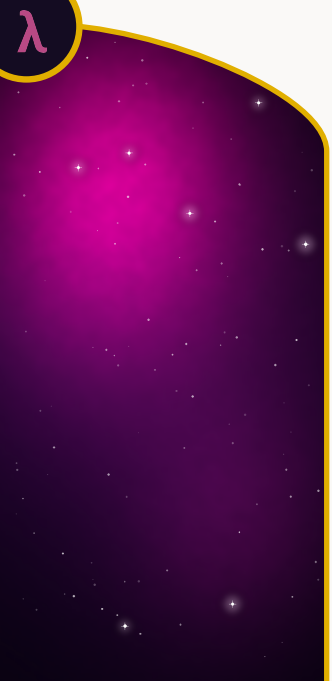
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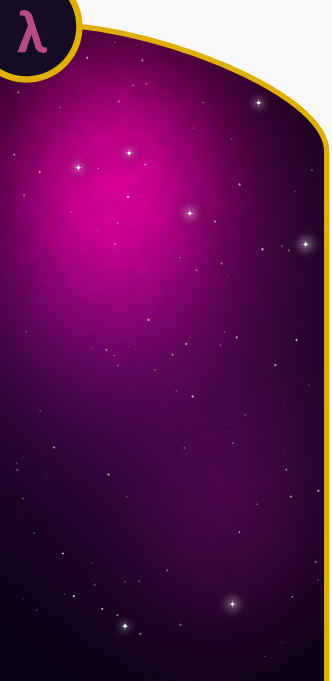
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- Each reachable pair (i, j) is computed at most once.
- Each computation performs $\Theta(1)$ work besides its recursive calls.
- Time: $\Theta(nk)$.
- Additional space: $\Theta(nk)$ for the table, plus recursion space.

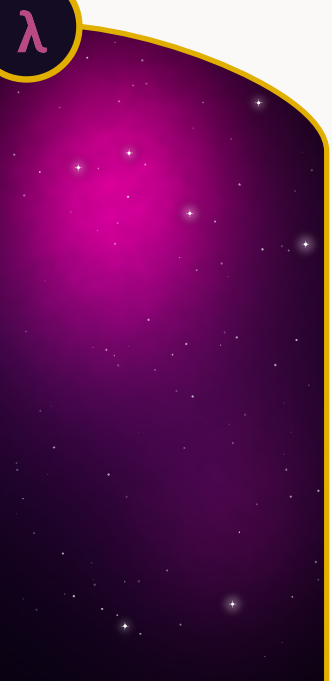


Summary



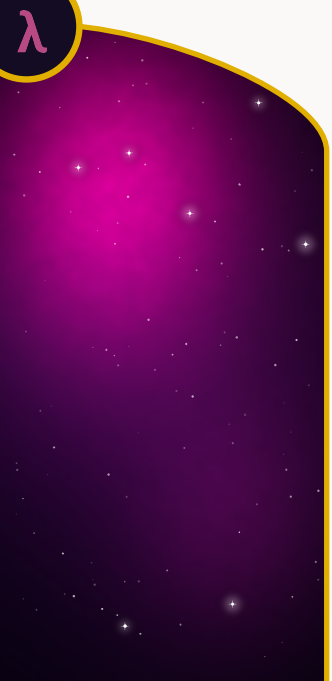
Summary

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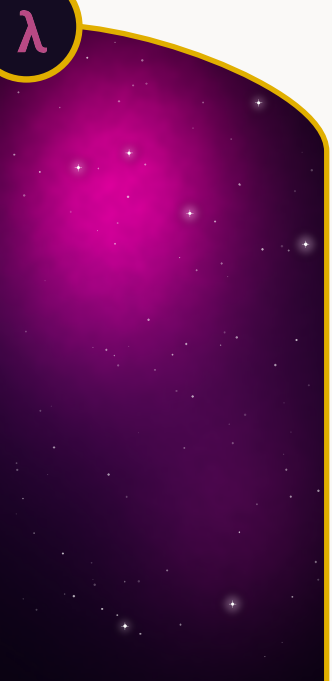
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- Binary search exploits sorted lists and random access to achieve $\Theta(\log n)$ time.
- Selection, insertion, and merge sort adapt familiar list-processing ideas to mutable lists.
- Memoization replaces repeated evaluation with table lookup.
- The table representation determines the cost of that lookup.